Build a complete prediction

LEARNING TARGET I can connect forces, velocity change, and final position.

A · KEY VOCABULARY

The chain	Net force → acceleration → velocity → position.
Constant net force	Gives constant acceleration: a straight v–t line.
Zero net force	Velocity stays the same; position still changes.
Range	A central value with limits. Not a guarantee.

B · KEY EQUATIONS

$$a = F_{net} \div m$$
$$v_f = v_i + a \times t$$
$$\Delta x = \tfrac{1}{2}(v_i + v_f) \times t$$
$$x_f = x_i + \Delta x$$

N, kg	force, mass
m/s², m/s, m	a, v, x

C · COLOR IT · ONE CHAIN, FOUR LINKS

2 kg cart, from rest at 1 m, for 2 s.

1 NET FORCE
6 − 2 = 4 N
pull minus friction

→

2 ACCEL.
4 ÷ 2 = 2 m/s²
$F_{net} \div m$

→

3 VELOCITY
0 + 2×2 = 4 m/s
gain 2 m/s each second

→

4 POSITION
1 + 4 = 5 m
Δx = 2 m/s avg × 2 s

Report the range: **5 ± 1 m → 4 to 6 m** – the central value is not guaranteed.

- ☐ **1 · ORANGE** Force
- ☐ **2 · GREEN** Acceleration (speeding up)
- ☐ **3 · PURPLE** Velocity
- ☐ **4 · BLUE** Position

D · WORKED EXAMPLE

A cart has zero net force and already moves at +2 m/s. After 3 s, find its velocity and displacement.

GIVEN	EQUATION	WORK	ANSWER
F_{net} = 0 N v = +2 m/s t = 3 s	a = 0 → v constant Δx = v × t	v stays +2 m/s Δx = 2 × 3	**+2 m/s, +6 m**

E · THE 2026-XJ THREAD

DAY 20 OF 22 · NEW TOOL: THE FULL CHAIN

The chain is complete: net force → acceleration → velocity → position. That is the whole route from the data on XJ to a forecast of where it will arrive.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Connect the forces to the final position

The complete chain from net force to a motion prediction

We have learned how forces change velocity and how velocity changes position. Today we connect those ideas in one calculation. I want you to keep the same object, positive direction, and time interval throughout the chain. Each result supplies information needed for the next step.

The diagram begins with a **2 kg cart** at rest at **1 m**. A **6 N pull** acts right while **2 N of friction** acts left. These forces stay constant for **2 s**. Choose right as positive. First find F_{net} = **6 N − 2 N = +4 N**. Then $a = F_{net} \div m$ = **4 N ÷ 2 kg = 2 m/s²**.

The cart gains 2 m/s of velocity each second. Use $v_f = v_i + at$ to find v_f = **0 m/s + (2 m/s²)(2 s) = +4 m/s**. Constant net force on a fixed mass gives constant acceleration, so its velocity-time graph would be a straight rising line.

Because the acceleration is constant, average velocity is **(0 + 4) m/s ÷ 2 = 2 m/s**. Multiply by the two seconds to find Δx = **(2 m/s)(2 s) = +4 m**. Now include the starting position: x_f = **1 m + 4 m = 5 m**. The cart's displacement is four meters, while its final position is the five-meter mark.

The diagram reports an illustrative bound of **5 ± 1 m**, meaning a range from **4 m to 6 m**. That one-meter bound is supplied for the example; it does not come from the force calculation alone. The center is not guaranteed to be the exact measured result, and the range is not a probability statement.

The worked example on the same page tests a different case: a cart already moving at **+2 m/s** has **zero net force** for **3 s**. Zero net force gives zero acceleration, so the velocity stays at +2 m/s. Its displacement is Δx = **(2 m/s)(3 s) = +6 m**. The cart keeps moving even though no unbalanced force acts on it.

Notice what information each step needs. Force and mass give acceleration. Starting velocity and time are needed to find final velocity. Starting position is needed to find final position. Orange identifies the force step, purple identifies velocity, and blue identifies position. Green in this particular diagram marks speeding up; acceleration itself does not always mean increasing speed.

For our fictional asteroid, 2026-XJ, this chain connects the observations and interactions to a forecast. A force calculation alone is not the whole prediction. We also need the existing motion and starting location, along with a model that fits the interval being studied.